A Method Based on Convex Cone Model for Image-Set Classification with CNN Features

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***Abstract*—In this paper, we propose a method for image-set classification based on convex cone models, focusing on the effectiveness of convolutional neural network (CNN) features as its input. CNN feature has non-negative values when using the rectified linear unit as an activation function. This naturally leads us to model a set of CNN features by a convex cone and measure the geometrical similarity of convex cones in classification. To achieve this framework, we define sequentially multiple angles between two convex cones by repeating the alternating least square method, and then define the geometrical similarity between the cones by using the obtained angles. Moreover, to enhance our method, we introduce a discriminant space, which maximizes the between-class variance (gaps) and minimizes the within-class variance of the projected convex cones onto the discriminant space, like Fisher discriminant analysis. Finally, the classification is conducted by measuring the similarity between projected convex cones. The effectiveness of the proposed method is demonstrated through evaluation experiments on a private database of a multi-view hand shape dataset, and two public databases.**

***Index Terms*—Image-set based method, Mutual convex cone method, Convex cone representation**

I. Introduction

In this paper, we propose a method for image-set classification based on convex cone models that can deal with various types of features with the non-negative constraint. Among them, we discuss on the effectiveness of the combination with convolutional neural network (CNN) features, which are extracted from a high-level hidden layer of a learned CNN.

For the last decade, image-set based classification methods [1]–[7], in particular, subspace based methods such as mutual subspace method (MSM) [1] and constrained MSM (CMSM) [2], [6], have been gaining much attention in various applications using multi-view images and videos, such as 3D object recognition and motion analysis, as they can handle with a set of images effectively. In these methods, a set of images is compactly represented by a subspace in high dimensional vector space, where the subspace is generated by applying PCA to the image set without data centering. The classification of an input subspace is conducted by using the canonical angles [8], [9] between the input and each reference subspace as the similarity.

The conventional subspace based methods assume a raw intensity vector or a hand-craft feature extracted as their input. Regarding more discriminant features, many studies have recently revealed that CNN features can efficiently work as inputs for various types of classifiers [10]–[13]. Inspired by these results, we employ CNN features as inputs for our image-set classification method based on convex cone models. Figure 1 illustrates the conceptual diagram of this basis idea.

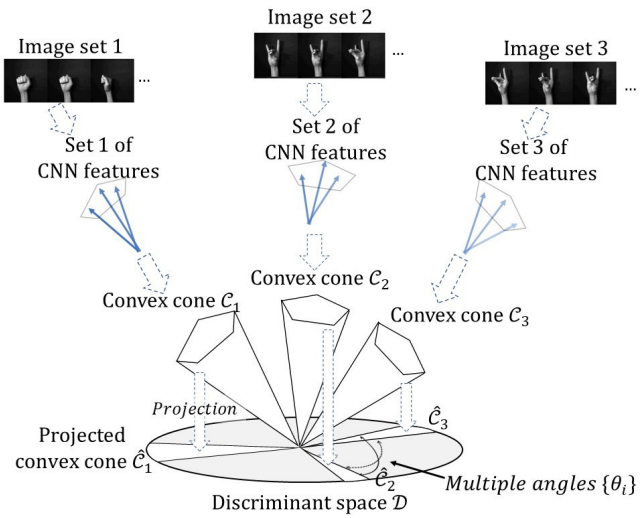

Fig. 1. Conceptual diagram of the proposed constrained mutual convex cone method (CMCM). Firstly, a set of CNN features is extracted from an image set. Then, each set of CNN features is represented by a convex cone. After the convex cones are projected onto the discriminant space $\mathcal{D}$, the classification is done by measuring a similarity with the angles $\{\theta_i\}$ between the two projected convex cones $\hat{\mathcal{C}}_i$ and $\hat{\mathcal{C}}_j$.

CNN feature vectors have only non-negative values when the rectified linear unit (ReLU) [14] is used as an activation function. This characteristic does not allow the combination of CNN features with negative coefficients and ensures that a set of CNN features forms a convex cone in high dimensional vector space as will be described in Sec.II-C. For example, it is well known that a set of front face images under various illuminations forms a convex cone, which is called illumination cone [15], [16]. Several previous works demonstrate the advantages of convex cone representation against subspace representation [17], [18]. These advantages naturally motivate us to replace a subspace with a convex cone in modeling a set of CNN features.

To achieve this framework, first of all, we need to consider how to calculate the geometrical similarity between two convex cones. To this end, we define multiple angles

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between two convex cones by reference to the definition of the canonical angles [8], [9] between two subspaces. Although it is known that the canonical angles between two subspaces can be analytically obtained from the orthonormal basis vectors of the two subspaces, the definition of angles between two convex cones is not trivial as we need to consider the non-negative constraint. In this paper, we define multiple angles between convex cones sequentially from the smallest one by repeatedly applying the alternating least square method [19]. Then, the geometrical similarity between two convex cones is defined with the obtained angles. We call a classification method using this similarity *mutual convex cone method* (MCM), corresponding to the mutual subspace method (MSM).

Moreover, to enhance the performance of MCM, we introduce a discriminant space $\mathcal{D}$, which maximizes the between-class variance (gap) among convex cones projected onto the discriminant space and minimizes the within-class variance of the projected convex cones, like Fisher discriminant analysis [20]. The class separability can be increased by projecting the class convex cones $\{\mathcal{C}_c\}$ onto the discriminant space $\mathcal{D}$, as shown in Fig.1. As a result, the classification ability of MCM is enhanced as well as in the projection of class subspaces onto a generalized difference subspace (GDS) in CMSM [6]. Finally, we perform the classification by using the angles between the projected convex cones $\{\hat{\mathcal{C}}_c\}$. We call this enhanced method "*constrained mutual convex cone method* (CMCM)", corresponding to constrained MSM (CMSM).

The main contributions of this paper are summarized as follows.

1) We introduce a convex cone representation to represent a set of CNN features accurately and compactly.
2) We introduce two novel mechanisms in our image-set based classification: a) multiple angles between two convex cones to measure the similarity between them, and b) a discriminant space to increase the class separability among class convex cones.
3) We propose two novel image-set based classification methods, called mutual convex cone method (MCM) and constrained mutual convex cone method (CMCM), based on convex cone representation and the discriminant space.

The rest of this paper is organized as follows. In Section 2, we describe the algorithms of conventional methods such as MSM and CMSM. In Section 3, we describe the details of the proposed method. In Section 4, we demonstrate the validity of the proposed method through classification experiments on a private database of multi-view hand shape and two public datasets, ETH-80 [21], and CMU face [22] datasets. Section 5 concludes the paper.

II. Related work

In this section, firstly, we describe the algorithms of the mutual subspace method (MSM) and the constrained MSM (CMSM), which are well known as one of the standard methods on the image-set classification. Then, we overview the concepts of a convex cone.

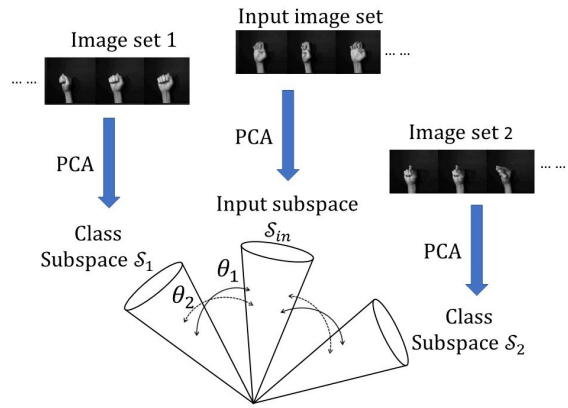

Fig. 2. Conceptual diagram of conventional MSM. Each image set is represented by a subspace, where it is generated by applying the PCA to the set. In the classification, the similarity between two subspaces is measured by using the canonical angles between them. An input subspace is classified to the class corresponding to the subspace with the highest similarity.

A. Mutual subspace method based on canonical angles

MSM is a classifier based on canonical angles between two subspaces, where each subspace represents an image set.

Given N_1-dimensional subspace $\mathcal{S}_1$ and N_2-dimensional subspace $\mathcal{S}_2$ in d-dimensional vector space, where $N_1 \leq N_2$, the canonical angles $\{0 \leq \theta_1, ..., \theta_{N_1} \leq \frac{\pi}{2}\}$ between the $\mathcal{S}_1$ and $\mathcal{S}_2$ are recursively defined as follows [8], [9]:

$$\cos\theta_i = \max_{\mathbf{u}\in\mathcal{S}_1}\max_{\mathbf{v}\in\mathcal{S}_2} \mathbf{u}^{\mathrm{T}}\mathbf{v} = \mathbf{u}_i^{\mathrm{T}}\mathbf{v}_i, \quad (1)$$
$$s.t. \|\mathbf{u}_i\|_2 = \|\mathbf{v}_i\|_2 = 1, \mathbf{u}_i^{\mathrm{T}}\mathbf{u}_j = \mathbf{v}_i^{\mathrm{T}}\mathbf{v}_j = 0, i \neq j,$$

where $\mathbf{u}_i$ and $\mathbf{v}_i$ are the canonical vectors forming the i-th smallest canonical angle θ_i between $\mathcal{S}_1$ and $\mathcal{S}_2$. The j-th canonical angle θ_j is the smallest angle in a direction orthogonal to the canonical angles $\{\theta_k\}_{k=1}^{j-1}$ as shown in Fig.3.

The canonical angles can be calculated from the orthogonal projection matrices onto subspaces $\mathcal{S}_1, \mathcal{S}_2$. Let $\{\mathbf{\Phi}_i\}_{i=1}^{N_1}$ be basis vectors of $\mathcal{S}_1$ and $\{\mathbf{\Psi}_i\}_{i=1}^{N_2}$ be basis vectors of $\mathcal{S}_2$. The projection matrices $\mathbf{P}_1$ and $\mathbf{P}_2$ are calculated as $\sum_{i=1}^{N_1} \mathbf{\Phi}_i{\mathbf{\Phi}_i}^{\mathrm{T}}$ and $\sum_{i=1}^{N_2} \mathbf{\Psi}_i{\mathbf{\Psi}_i}^{\mathrm{T}}$, respectively. $\cos^2\theta_i$ is the i-th largest eigenvalue of $\mathbf{P}_1^{\mathrm{T}}\mathbf{P}_2$ or $\mathbf{P}_2^{\mathrm{T}}\mathbf{P}_1$. Alternatively, the canonical angles can be easily obtained by applying the SVD to the orthonormal basis vectors of the subspaces.

The geometrical similarity between two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$ is defined by using the canonical angles as follows:

$$sim(\mathcal{S}_1, \mathcal{S}_2) = \frac{1}{N_1}\sum_{i=1}^{N_1}\cos^2\theta_i. \quad (2)$$

In MSM, an input subspace $\mathcal{S}_{in}$ is classified by comparing it with class subspaces $\{\mathcal{S}_c\}_{c=1}^{C}$ using this similarity.

B. Constrained MSM

The essence of constrained MSM (CMSM) is to perform the MSM on a generalize difference subspace (GDS) [6] as shown in Fig.4. GDS is designed to contain only difference

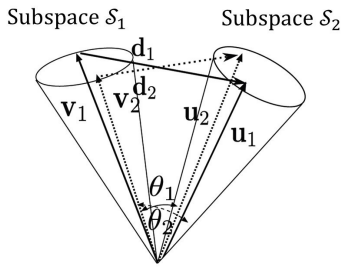

Fig. 3. Conceptual diagram of the canonical angles and canonical vectors. 1-st canonical vectors $\mathbf{u}_1, \mathbf{v}_1$ form the smallest angle θ_1 between the subspaces. 2-nd canonical vectors $\mathbf{u}_2, \mathbf{v}_2$ form the smallest angle θ_2 in orthogonal direction to θ_1.

components among subspaces $\{\mathcal{S}_c\}_{c=1}^{C}$. Thus, the projection of class subspaces onto GDS can increase the class separability among the class subspaces to improve largely the classification ability of MSM [6].

C. Convex cone model

In this subsection, we explain the definition of a convex cone and the projection a vector onto a convex cone. A convex cone $\mathcal{C}$ is defined by using finite basis vectors $\{\mathbf{b}_i\}_{i=1}^{r}$ as follows:

$$\{\mathbf{a} \in \mathcal{C} | \mathbf{a} = \sum_{i=1}^{r} w_i \mathbf{b}_i, w_i \geq 0\}. \tag{3}$$

As can be seen from this definition, the difference between the concepts of a subspace and a convex cones is whether there are non-negative constraints on the combination coefficients w_i or not.

Given a set of feature vectors $\{\mathbf{f}_i\}_{i=1}^{N} \in \mathbb{R}^d$, the basis vectors $\{\mathbf{b}_i\}_{i=1}^{r}$ of a convex cone representing a distribution of $\{\mathbf{f}_i\}$ can be obtained by the non-negative matrix factorization (NMF) [23], [24]. Let $\mathbf{F} = [\mathbf{f}_1 \mathbf{f}_2 ... \mathbf{f}_N] \in \mathbb{R}^{d \times N}$ and $\mathbf{B} = [\mathbf{b}_1 \mathbf{b}_2 ... \mathbf{b}_r] \in \mathbb{R}^{d \times r}$. NMF generates the basis vectors $\mathbf{B}$ by solving the following optimization problem:

$$\underset{\mathbf{B},\mathbf{W}}{\arg\min} \|\mathbf{F} - \mathbf{B}\mathbf{W}\|_F \quad s.t. \quad (\mathbf{B})_{i,j}, (\mathbf{W})_{i,j} \geq 0, \tag{4}$$

where $\|\cdot\|_F$ denotes the frobenius norm. We use the alternating non-negativity-constrained least squares based method [24] to solve this problem.

Although the basis vectors can be easily obtained by the NMF, the projection a vector onto the convex cone is slightly complicated due to the non-negative constraint on the coefficients. In [17], the projection of a vector $\mathbf{x}$ onto the convex cone is conducted by solving the non-negative least square method [25] as follows:

$$\underset{\{w_i\}}{\arg\min} \|\mathbf{x} - \sum_{i=1}^{r} w_i \mathbf{b}_i\|_2 \quad s.t. \quad w_i \geq 0. \tag{5}$$

The projected vector $\hat{\mathbf{x}}$ is obtained as $\hat{\mathbf{x}} = \sum_{i=1}^{r} w_i \mathbf{b}_i$.

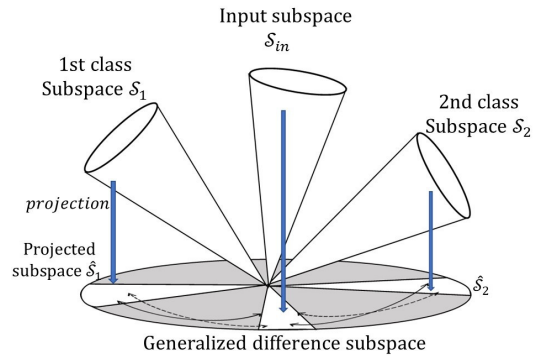

Fig. 4. Conceptual diagram of constrained MSM (CMSM). By projecting class subspaces onto the generalized difference subspace, the separability between the classes are increased. Then, by measuring the similarities among the projected subspaces using the canonical angles, the input subspace is classified into either of class 1 and 2.

In the end, the angle θ between the convex cone and a vector $\mathbf{x}$ can be calculated as follows:

$$\cos\theta = \frac{\mathbf{x}^{\mathrm{T}} \hat{\mathbf{x}}}{\|\mathbf{x}\|_2 \|\hat{\mathbf{x}}\|_2}. \tag{6}$$

III. Proposed method

In this section, we explain the algorithms of mutual convex cone method (MCM) and constrained mutual convex cone method (CMCM), after providing the definition of a geometrical similarity between two convex cones.

A. Geometrical similarity between two convex cones

We define the geometrical similarity between two convex cones. To this end, we consider how to define multiple angles between two convex cones. Given two convex cones $\mathcal{C}_1$ and $\mathcal{C}_2$ which are formed by basis vectors $\{\mathbf{b}_i^1\}_{i=1}^{N_1} \in \mathbb{R}^d$ and $\{\mathbf{b}_i^2\}_{i=1}^{N_2} \in \mathbb{R}^d$, respectively. Assume that $N_1 \leq N_2$ for convenience sake. Angles between two convex cones cannot be obtained analytically like the canonical angles between two subspaces, as we need to consider the non-negative constraint. Alternatively, we find two vectors, $\mathbf{p} \in \mathcal{C}_1$ and $\mathbf{q} \in \mathcal{C}_2$, which are the closest to each other. Then, we define the angle between the two convex cones as the one formed by the two vectors. In this way, we sequentially define multiple angles from the smallest one in order.

First of all, we search a pair of d-dimensional vectors $\mathbf{p}_1 \in \mathcal{C}_1$ and $\mathbf{q}_1 \in \mathcal{C}_2$, which have the maximum correlation by using the alternating least square method (ALS) [19]. The first angle θ_1 is defined as the angle formed by $\mathbf{p}_1$ and $\mathbf{q}_1$. The pair of $\mathbf{p}_1$ and $\mathbf{q}_1$ can be found by using the following algorithm:

Algorithm for searching the pair of $\mathbf{p}_1$ and $\mathbf{q}_1$

Let $\mathcal{P}_1\mathbf{y}$ and $\mathcal{P}_2\mathbf{y}$ be the projections of a vector $\mathbf{y}$ onto $\mathcal{C}_1$ and $\mathcal{C}_2$, respectively. For the details of the projection, see the Section II-C.

1) Randomly initialize $\mathbf{y} \in \mathbb{R}^{\mathbf{d}}$.
2) $\mathbf{p}_1 = \mathcal{P}_1\mathbf{y} / \|\mathcal{P}_1\mathbf{y}\|_2$.

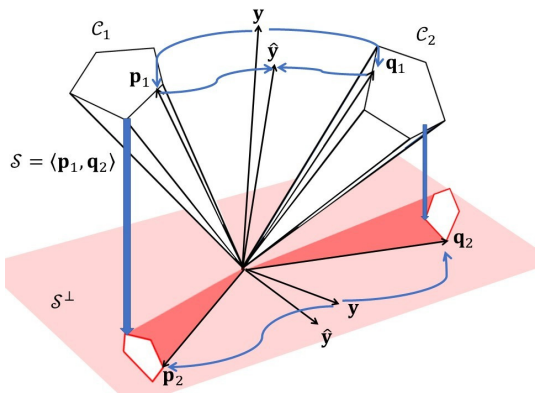

Fig. 5. Conceptual diagram of the procedure for searching pairs of vector $\{\mathbf{p}_i, \mathbf{q}_i\}$. The first pair of $\mathbf{p}_1$ and $\mathbf{q}_1$ can be found by the alternating least square method. After that, the second pair of $\mathbf{p}_2$ and $\mathbf{q}_2$ is obtained by searching on the orthogonal complement space $\mathcal{S}^{\perp}$ of $\mathcal{S} =< \mathbf{p}_1, \mathbf{q}_1 >$.

3) $\mathbf{q}_1 = \mathcal{P}_2\mathbf{y}/\|\mathcal{P}_2\mathbf{y}\|_2$.
4) $\hat{\mathbf{y}} = (\mathbf{p}_1 + \mathbf{q}_1)/2$.
5) If $\|\hat{\mathbf{y}} - \mathbf{y}\|_2$ is sufficiently small, the procedure is completed. Otherwise, return to 2) as $\mathbf{y} = \hat{\mathbf{y}}$.
6) $\cos^2\theta_1 = (\frac{\mathbf{p}_1^{\mathrm{T}}\mathbf{q}_1}{\|\mathbf{p}_1\|_2\|\mathbf{q}_1\|_2})^2$.

For the second angle θ_2, we search a pair of vectors $\mathbf{p}_2$ and $\mathbf{q}_2$ with the maximum correlation, while they have the minimum correlation with $\mathbf{p}_1$ and $\mathbf{q}_1$ at the same time. Such a pair can be found by applying ALS to the projected convex cones $\mathcal{C}_1$ and $\mathcal{C}_2$ on the orthogonal complement space $\mathcal{S}^{\perp}$ of the subspace $\mathcal{S}$ spanned by the vectors $\mathbf{p}_1$ and $\mathbf{q}_1$ as shown in Fig.5. θ_2 is formed by $\mathbf{p}_2$ and $\mathbf{q}_2$. In this way, we can obtain all the pairs of vector $\mathbf{p}_i, \mathbf{q}_i$ forming i-th angle θ_i, $(i = 1, \ldots, N_1)$.

With these obtained angles $\{\theta_i\}_{i=1}^{N_1}$, we define a geometrical similarity sim between two convex cones $\mathcal{C}_1$ and $\mathcal{C}_2$ as follows:

$$sim(\mathcal{C}_1, \mathcal{C}_2) = \frac{1}{N_1}\sum_{i=1}^{N_1}\cos^2\theta_i. \tag{7}$$

B. Mutual convex cone method

Mutual convex cone method (MCM) classifies an input convex cone based on the similarities defined by Eq.(7) between the input and class convex cones. MCM consists of two phases, a training phase and a recognition phase as shown in Fig.6.

Given C class sets with L images$\{\mathbf{x}_i^c\}_{i=1}^{L}$.

Training Phase

1) CNN features $\{\mathbf{f}_i^c\}$ are extracted from the images $\{\mathbf{x}_i^c\}$ of class c. Then, the extracted CNN features are normalized as $\{\mathbf{f}_i^c/\|\mathbf{f}_i^c\|_2\}$.
2) The basis vectors of c class convex cone $\{\mathbf{b}_j^c\}$ are generated by applying NMF to the set of normalized CNN features.
3) $\{\mathbf{b}_j^c\}$ are registered as the reference convex cone of class c.
4) The above process is conducted for all C classes.

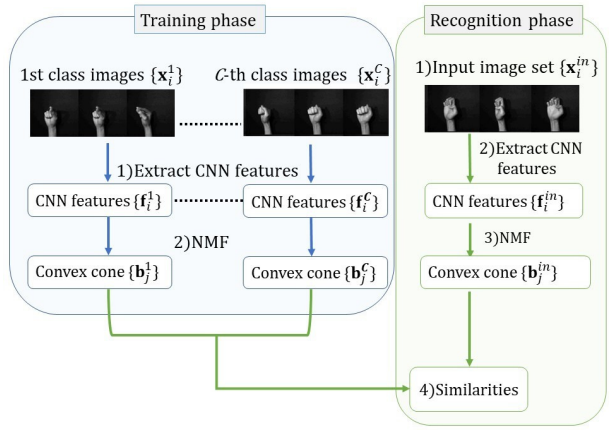

Fig. 6. The process flow of the proposed mutual convex cone method (MCM), which consists of a training phase and a testing phase.

Recognition Phase

1) A set of images $\{\mathbf{x}_i^{in}\}$ is inputted.
2) CNN features $\{\mathbf{f}_i^{in}\}$ are extracted from the images $\{\mathbf{x}_i^{in}\}$. Then, the extracted CNN features are normalized as $\{\mathbf{f}_i^{in}/\|\mathbf{f}_i^{in}\|_2\}$.
3) The basis vectors of the input convex cone $\{\mathbf{b}_j^{in}\}$ are generated by applying NMF to the input set of normalized CNN features.
4) The input image set $\{\mathbf{x}_i^{in}\}$ is classified based on the similarity (Eq.(7)) between the input convex cone $\{\mathbf{b}_j^{in}\}$ and each c-th class reference convex cone $\{\mathbf{b}_j^c\}$.

C. Generation of discriminant space

To enhance the class separability among multiple class convex cones, we introduce a discriminant space $\mathcal{D}$, which maximizes between-class variance $\mathbf{S_b}$ and minimizes within-class variance $\mathbf{S_w}$ regarding the convex cones projected on $\mathcal{D}$, like Fisher discriminant analysis (FDA). In our method, the between class variance is replaced with gaps among convex cones. We define these gaps as follows. Let $\mathcal{C}_c$ be c-th class convex cone with basis vectors $\{\mathbf{b}_i^c\}_{i=1}^{N_c}$, $\mathcal{P}_c$ be the projection operation of a vector onto $\mathcal{C}_c$, and C be the number of the classes. We consider C vectors $\{\mathbf{p}_1^c\}$, $(c = 1, 2, ..., C)$ such that the sum of the correlation $\sum_{c\neq c'}(\mathbf{p}_1^c)^{\mathrm{T}}\mathbf{p}_1^{c'}/(\|\mathbf{p}_1^c\|_2\|\mathbf{p}_1^{c'}\|_2)$ is maximum. Such a set of vectors can be obtained by using the following algorithm. This algorithm is almost the same as the generalized canonical correlation analysis (CCA) [26], [27] except that the non-negative least squares (LS) method is used instead of the standard LS method.

Procedure to search a set of first vectors $\{\mathbf{p}_1^c\}_{c=1}^{C}$

1) Randomly initialize $\mathbf{y}_1$.
2) Project $\mathbf{y}_1$ onto each convex cone, and then normalize the projection as $\mathbf{p}_1^c = \mathcal{P}_c\mathbf{y}_1/\|\mathcal{P}_c\mathbf{y}_1\|_2$.
3) $\hat{\mathbf{y}}_1 = \sum_{c=1}^{C}\mathbf{p}_1^c/C$.
4) If $\|\mathbf{y}_1 - \hat{\mathbf{y}}_1\|_2$ is sufficiently small, the procedure is completed. Otherwise, return to 2) as $\mathbf{y}_1 = \hat{\mathbf{y}}_1$.

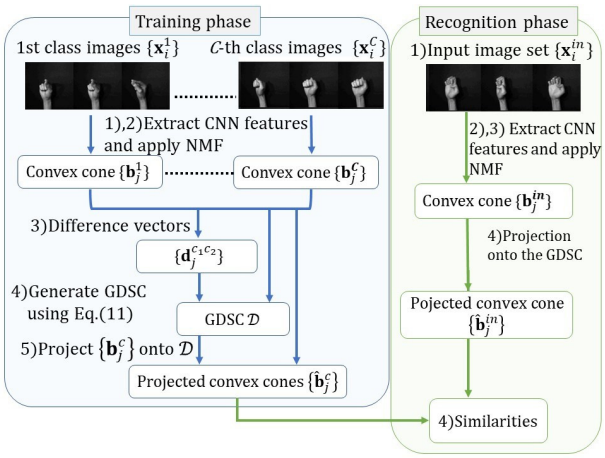

Fig. 7. The process flow of the proposed constrained MCM (CMCM). CMCM is a enhanced version of MSM with the projection of class subspaces onto the discriminant space $\mathcal{D}$.

Next, we search a set of the second vectors $\{\mathbf{p}_2^c\}$ with the maximum sum of the correlations under the constraint condition that they have the minimum correlation with the previously found $\{\mathbf{p}_1^c\}$. The second vectors $\{\mathbf{p}_2^c\}$ can be obtained by performing the above procedure on the orthogonal complement space of the vector $\mathbf{y}_1$. In the following, a set of j-th vectors $\{\mathbf{p}_j^c\}$ can be sequentially obtained by conducting the same procedure on the orthogonal complement space of $\{\mathbf{y}_k\}_{k=1}^{j-1}$. In this way, we finally obtain the sets of $\{\mathbf{p}_j^c\}$. With the obtained sets of $\{\mathbf{p}_j^c\}$, we define a difference vector $\{\mathbf{d}_j^{c_1c_2}\}$ as follows:

$$\mathbf{d}_j^{c_1c_2} = \mathbf{p}_j^{c_1} - \mathbf{p}_j^{c_2}. \tag{8}$$

Considering that each difference vector represents the gap between the two convex cones, we define $\mathbf{S_b}$ with them as follows:

$$\mathbf{S_b} = \sum_{j=1}^{\min(\{N_c\})} \sum_{c_1=1}^{C-1} \sum_{c_2=c_1+1}^{C} \mathbf{d}_j^{c_1c_2}(\mathbf{d}_j^{c_1c_2})^{\mathrm{T}}. \tag{9}$$

Next, we define the within-class variance $\mathbf{S_w}$ using the basis vectors $\{\mathbf{b}_i^c\}$ of all the class convex cones as follows:

$$\mathbf{S_w} = \sum_{c=1}^{C} \sum_{i=1}^{N_c} (\mathbf{b}_i^c - \mu_c)(\mathbf{b}_i^c - \mu_c)^{\mathrm{T}}, \tag{10}$$

where $\mu_c = \sum_{i=1}^{N_c} \mathbf{b}_i^c / N_c$. Finally, the N_d-dimensional discriminant space $\mathcal{D}$ is spanned by N_d eigenvectors $\{\phi_i\}_{i=1}^{N_d}$ corresponding to the N_d largest eigenvalues $\{\gamma_i\}_{i=1}^{N_d}$ of the following eigenvalue problem:

$$\mathbf{S_b}\phi_i = \gamma_i \mathbf{S_w}\phi_i. \tag{11}$$

D. Constrained mutual convex cone method

We construct constrained MCM (CMCM) by incorporating the projection onto the discriminant space $\mathcal{D}$ into the MCM. CMCM consists of a training phase and a testing phase as shown in Fig.7. In the following, we explain each phase in the case that C classes have L images$\{\mathbf{x}_i^c\}_{i=1}^L$.

Training Phase

1) CNN features $\{\mathbf{f}_i^c\}$ are extracted from the images $\{\mathbf{x}_i^c\}$. Further, the extracted CNN features are normalized as $\{\mathbf{f}_i^c / \|\mathbf{f}_i^c\|_2\}$.
2) The basis vectors of c class convex cone $\{\mathbf{b}_j^c\}$ are generated by applying NMF to each class set of normalized CNN features.
3) Sets of difference vectors $\{\mathbf{d}_j^{c_1c_2}\}$ are generated by conducting the procedure described in the section III-C.
4) The discriminant space $\mathcal{D}$ are generated by solving Eq.(11) using $\{\mathbf{b}_j^c\}$ and $\{\mathbf{d}_j^{c_1c_2}\}$.
5) The basis vectors $\{\mathbf{b}_j^c\}$ are projected onto the discriminant space $\mathcal{D}$ and then the lengths of the projected basis vectors are normalized to 1.0. A set of these basis vectors $\{\hat{\mathbf{b}}_j^c\}$ forms the projected convex cone.
6) $\{\hat{\mathbf{b}}_j^c\}$ are registered as the reference convex cone of class c.

Recognition Phase

1) A set of images $\{\mathbf{x}_i^{in}\}$ is inputted.
2) CNN features $\{\mathbf{f}_i^{in}\}$ are extracted from the images $\{\mathbf{x}_i^{in}\}$. Further, the extracted CNN features are normalized as $\{\mathbf{f}_i^{in} / \|\mathbf{f}_i^{in}\|_2\}$
3) The basis vectors of a convex cone $\{\mathbf{b}_j^{in}\}$ are generated by applying NMF to the set of the normalized CNN features.
4) The basis vectors $\{\mathbf{b}_j^{in}\}$ are projected onto the discriminant space $\mathcal{D}$ and then the lengths of the projected basis vectors are normalized to 1.0. The normalized projections are represented by $\{\hat{\mathbf{b}}_j^{in}\}$
5) The input set $\{\mathbf{x}_i^{in}\}$ is classified based on the similarity (Eq.(7)) between the input convex cone $\{\hat{\mathbf{b}}_j^{in}\}$ and each class reference convex cone $\{\hat{\mathbf{b}}_j^c\}$.

IV. Evaluation Experiments

In this section, we demonstrate the effectiveness of the proposed methods through four experiments using three datasets: 1) multi-view hand shape dataset [28], 2) ETH-80 dataset [21], 3) CMU Multi-PIE face dataset [22]. In the first three experiments, we conducted the classification on the three datasets in the case that the number of training samples is sufficiently large. In contrast, in the final experiment, we show the robustness of the proposed methods against small sample size (SSS), considering the situation that only few training samples can be used to learn the networks. For the implementation of the methods, we used the NMF toolbox [29] and keras [30].

A. Hand shape classification

1) Details of the dataset: The multi-view hand shape dataset consists of 30 classes of hand shapes. Each class was taken from 100 subjects at the speed of 1 fps for 4 seconds by using a multi-camera system, which is equipped with seven

Fig. 8. Sample images of the multi-view hand shape dataset which is used in the experiments. Each row shows a hand shape taken from various viewpoints.

TABLE I
EXPERIMENT RESULT FOR THE HAND SHAPE DATASET.

Feature	Method	Error Rate(%)
Raw image	MSM [6]	22.55
	CMSM [6]	17.12
	KMSM [6]	12.52
	KCMSM [6]	9.28
	KOMSM [6]	9.43
FFT feature	MSM [6]	17.69
	CMSM [6]	8.78
	KMSM [6]	7.13
	KCMSM [6]	5.84
	KOMSM [6]	5.66
CNN feature	softmax	1.11
	MSM	1.06
	CMSM	0.97
CNN feature	MCM	1.47
	CMCM	**0.92**

synchronized cameras intervals of 10 degrees. During the data collection, the subjects were asked to rotate their hands at a constant speed to increase the number of viewpoints. Figure 8 shows several sample images of the dataset. The total number of the collected images is 84000 (= 30 classes$\times$4 frames$\times$7 cameras $\times$100 subjects).

2) Experimental protocol: We used the same protocol as in [6]. We divided randomly all the subjects into two sets. One set was used for training, and the other was used for testing. That is, a reference convex cone of each hand shape was generated from the set of 1,400 (=7 cameras$\times$4 frames$\times$50 subjects) images. As an input image set, we used 28 (=7 cameras$\times$4 frames) images. The total number of convex cones used for testing is 1,500 (=30 shapes$\times$50 subjects). We evaluated the classification performance of each method in terms of the average error rate (ER) of ten trials using randomly divided datasets.

We decided the parameters for the methods by the cross validation using the training data. For MSM and CMSM with CNN features, the dimensions of class and input subspaces, and GDS were set to 80, 5, and 200, respectively. For the conventional methods with raw images and FFT features, we used the same parameters as [6]. For MCM and CMCM, the numbers of the basis vectors of class and input convex cones were set to 30 and 7, respectively. The dimension N_d of the discriminant space $\mathcal{D}$ was set to 750.

To obtain CNN features under our experimental setting, we modified slightly the original ResNet-50 [31] trained by the Imagenet database [32] for our experiment conditions. First, we replaced the final 1000-way fully connected (FC) layer of the original ResNet-50 with a 1024-way FC layer with ReLU function. Further, we added a $class\ number$-way FC layer with softmax behind the previous 1024-way FC layer.

Moreover, to extract more effective CNN feature from our modified ResNet, we fine-tuned our ResNet by using the learning set. A CNN feature vector is extracted from the 1024-way FC layer every time when an image is input into our ResNet. As a result, the dimensionality d of a CNN feature vector is 1024.

In our fine-tuned CNN, an input image set is classified based on an average value of the output conviction degrees of each class from the last FC layer with softmax.

3) Hand shape classification result: Table I shows the average error rates of the different methods including the proposed method. In the table, KCMSM indicates a non-linear extension of CMSM by using kernel trick [33].

We can see that the subspace or convex cone based methods with CNN features are significantly superior to that with conventional features. We can confirm again the validity of CNN features. The results also support that a set of CNN features is more informative than the average value of the outputs from the last softmax layer. When comparing the convex cone based methods with the subspace based methods, CMCM achieves the best performance. This advantage suggests that a convex cone model is more suitable than a subspace model to represent a set of CNN features compactly and compare two sets effectively.

B. Object classification experiment

We conducted the experiment of object classification on the ETH-80 dataset [21].

1) Details of the ETH-80 and experimental protocol: The ETH-80 dataset consists of object images of 8 different categories, captured from 41 viewpoints. Each category has 10 kinds of object.

Five objects randomly sampled from each category set were used for training, and the remaining five objects were used for testing. As an input image set, we used 41 images for each object. We evaluated the classification performance of each method in terms of the average error rate (ER) of five trials using randomly divided datasets.

For MSM and CMSM, the dimensions of class subspaces, the input subspaces, and GDS were set to 55, 10, and 30, respectively. For MCM and CMCM, the numbers of the basis vectors of class and input convex cones were set to 30 and 7, respectively. The dimension N_d of the discriminant space $\mathcal{D}$ was set to 85. We decided these dimensionality by the cross-validation using the training data. CNN features were extracted from the fine-tuned ResNet under this experimental

TABLE II
EXPERIMENT RESULT FOR THE ETH-80 DATASET.

Feature	CNN feature				
Method	softmax	MSM	CMSM	MCM	CMCM
Error Rate(%)	10.50	12.00	11.00	9.50	**7.50**

TABLE III
EXPERIMENT RESULT FOR THE CMU PIE DATASET.

Feature	CNN feature				
Method	softmax	MSM	CMSM	MCM	CMCM
Error Rate(%)	4.22	4.01	3.75	4.84	**3.55**

setting, according to the same procedure as in the previous experiments.

2) Object classification result: Table II shows the error rates of the different methods. The CMCM achieve the highest accuracy. This result also supports that a convex cone model is appropriate to represent a set of CNN features compared with a subspace. In addition, we can confirm that the projection of the convex cones onto the discriminant space works well as a valid feature extraction.

C. Face classification experiment

We conducted the experiment of face classification on the CMU Multi-PIE dataset [22].

1) Details of the CMU dataset and experimental protocol: The CMU Multi-PIE dataset consists of face images of 337 different subjects captured from 15 viewpoints with 20 lighting conditions in 4 recording sessions. In this experiment, we used images of 129 subjects captured from three viewpoints: front, left and right. Thus, the total number of the images used for this experiment is 30960 (=129 subjects×3 views×20 illuminations×4 sessions).

Two sessions were used for training, and the remaining two sessions were used for testing. As an input image set, we used 10 images randomly sampled from an image set of each subject. For MSM and CMSM, the dimensions of class and input subspaces, and GDS were set to 20, 5, and 520, respectively. For MCM and CMCM, the numbers of the basis vectors of class and input convex cones were set to 20 and 5, respectively. The dimension N_d of the discriminant space $\mathcal{D}$ was set to 530. We decided these parameters by the cross validation. We used CNN features which were extracted from the fine-tuned ResNet by using the training data, following the experimental setting.

2) Face classification result: Table III shows the error rates of the different methods. From this, we can see that the CMCM achieves the highest performance, while the performance of the MCM is lowest. This result shows again the validity of the projection onto the discriminant space as a feature extraction. This implies that the gaps between convex cones captures useful geometrical information to enhance the class separability among all the class convex cones.

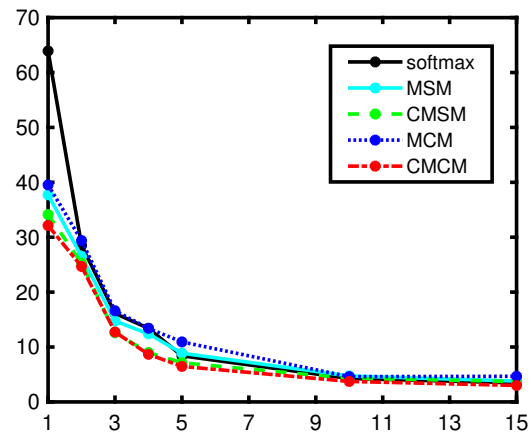

Fig. 9. Change in the error rates (%) against the number of training subjects. The horizontal axis denotes the number of training subjects and the vertical axis denotes the error rates.

D. Robustness against few training data

It is known as a serious problem that deep neural networks requires a large amount of training samples to accurately learn the networks. Therefore, the robustness against small sample size (SSS) is one of the necessary characteristic for the methods using CNN feature in practical applications. In this experiment, we evaluated the robustness of the different methods against the SSS.

1) Experimental protocol: In this experiment, we used again the hand shape dataset described in section IV-A1. The dataset was divided into two sets in the same way as the previous experiment. One set was used for training and another was used for testing. We evaluated the performances of the methods by setting the number of the subjects used for training to 1, 2, 3, 4, 5, 10, and 15. In each case, the total number of training images is 30 classes×7 cameras×4 frames× n subjects, ($n = 1, 2, 3, 4, 5, 10, 15$). As an input image set, we used 28 (=7 cameras ×4 frames) images as in the previous experiment. Thus, the total number of convex cones for testing is 1500 (=30 classes×50 subjects).

The parameters for the methods were decided by the cross validation using training images. For MSM and CMSM, the dimensions of class and input subspaces and GDS were set to 25, 7, and 725, respectively. For MCM and CMCM, the numbers of the basis vectors of class and input convex cones were set to 30 and 7, respectively. The dimension N_d of the discriminant space $\mathcal{D}$ was set to 800.

To extract CNN features from the images, we used the fine-tuned ResNet by using the training images under this experiment conditions.

2) Experiment result: Figure 9 shows the error rates in terms of the number N of training subjects. From the figure, we can see that overall the performance of CMCM is better than the other methods. In particular, CMCM works well when the number of training subjects N is small. For example, when N is 1, CMSM and CMCM achieve about a half of

the error rate of the softmax. Moreover, CMCM improves the performances of the subspace-based methods, MSM and CMSM. This supports that the convex cone can represent the distribution of a set of CNN features more accurately than the subspace-based methods.

V. Conclusion

In this paper, we proposed a method based on convex cone model for image-set classification, which is called constrained mutual convex cone method (CMCM). We discussed the combination of the proposed method and CNN features, though our method can be applied to various types of features with non-negative constraint.

The main contributions of this paper are 1) the introduction of a convex cone model to represent a set of CNN features compactly and accurately, 2) the definition of the geometrical similarity of two convex cones based on the angles between them, which are obtained by the alternating least square method, 3) the proposal of a method, which is named MCM, for classifying convex cones by using the angles as the similarity, 4) the introduction of a discriminant space, which maximizes between-class variance (gaps) between convex cones and minimizes within-class variance, 5) the proposal of constrained MCM (CMCM), which has been constructed by incorporating the above projection into MCM.

We demonstrated the validity of our methods through three experiments on the multi-view hand shape dataset, the public database, CMU PIE dataset and ETH-80. As a future work, we consider the possibility of introducing non-linear mapping by kernel function into the proposed methods.

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